

No. of Functional Features Included in the Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	MFA / SSO Authentication	Secure login for bank agents using Multi-Factor Auth and RBAC.	Security Module	Done	-
2	Agent Dashboard UI	Interactive interface for inputting applicant data and viewing results.	Frontend (React)	Done	-
3	Data Preprocessing	Automated pipeline handling missing values, encoding, and scaling.	Backend (ML)	Done	-
4	XGBoost ML Classification	Core predictive model determining credit card approval probability.	Backend (ML)	Done	-
5	Explainable AI (SHAP)	Generates transparent feature importance reports for every decision.	Backend (XAI)	Done	-
6	Applicant Data Storage	Securely logs applicant records, prediction results, and SHAP data.	Database	Done	-

7	Borderline Flagging	Flags 45-55% probability predictions for manual agent review.	Business Logic	Done	-
S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
8	Data Drift Monitoring	Alerts admins if input data distributions deviate significantly over time.	Monitoring	Pending	-

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core/Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Agent Dashboard UI
2	Backend/Logic	4	Data Preprocessing, XGBoost ML Classification, Explainable AI (SHAP), Borderline Flagging, Data Drift Monitoring
3	Database / Storage	1	Applicant Data Storage
4	API / Integration	1	External Credit Bureau API Integration (handled via FastAPI endpoints)
5	Security / Authentication	1	MFA / SSO Authentication